Incompatible formulae for circular-ensemble overlaps

Conjecture 00000001865

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The conjecture simultaneously asserts a mean

$$M(N, \beta, i) = N^{-2+2\beta}\{i(N-i)\}^{\beta-1}$$

and its specialization $M(N, 2, i) = N^{-2}$. These two assertions already contradict each other on their stated domain.

Theorem 1. *No mean function can satisfy both formulae for every $N \geq 2$, $\beta > 0$, and $1 \leq i < N$.*

Proof. Set $N = 2$, $\beta = 2$, and $i = 1$. The first formula gives $M = 2^2 \cdot 1 = 4$, whereas the claimed specialization gives $M = 2^{-2} = 1/4$. Since $4 \neq 1/4$, their conjunction is false. This argument needs no assumptions about the distribution of the matrices. □

There is a separate problem with the stated statistic. Distinct members v_i, v_j of an orthonormal eigenbasis have $\langle v_i, v_j \rangle = 0$. Consequently the expectation of $|\langle v_i, v_j \rangle|^2$ is zero under any probability measure on such families. In particular, the overlap of two vectors of the same orthonormal eigenbasis is not the overlap of two independent eigenbases. The circular unitary ensemble at $\beta = 2$ consists of unitary matrices and admits orthonormal eigenbases.

Lean correspondence. `predicted` uses real powers and is exactly the displayed expression. `conjecture_false` negates the conjunction for an arbitrary mean function on the full stated domain by specializing to the admissible triple above. `mean_overlap_zero` additionally proves the zero integral for every measure and every samplewise orthonormal family, using Mathlib's inner product and Bochner integral. No probability model is substituted for the ensemble in the main contradiction.

Source for terminology. Cunden, Mezzadri and O'Connell, Free Fermions and the Classical Compact Groups, Section 3, defines CUE using Haar measure on the unitary group. The disproof itself is the incompatible specialization, independently of that reference.

Verification. Lean 4.19.0 with Mathlib pinned to `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The accompanying project and verification transcript give the exact theorem and axiom audit. No proof placeholders, native evaluator proof bypass, or custom axiom declarations are used. This is a prepared submission; official acceptance has not been claimed.